

INDEPENDENT PEER-REVIEW PACKET

EC-013

Conditional Elder-Rule Defect Dichotomy

Cold Review Packet 8 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	8 of 20
Object ID	EC-013
Object class	Deterministic topology / Morse theory lemma
Exact statement	Under the declared Morse, distinct-critical-value, and Morse-Smale hypotheses, $\{D(M) \neq S\}$ is contained in Π_r union Γ_r .
Internal terminal label	CLOSED - EXACT CONDITIONAL ELDER-RULE DEFECT DICHOTOMY
Terminal date	2026-07-21
Recommended expertise	Morse theory / persistent homology / merge trees

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Deterministic pairing proof from the core pairing theorem; second-line reconstruction and terminal package.
Prior review record	A strong dependent verification checked the exact inclusion without importing probability estimates.
Binding exclusions	No probability, rate, transversality, adjacency-to-one, or P0.2 conclusion.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No probability, rate, transversality, adjacency-to-one, or P0.2 conclusion.

4. Object-specific hostile checks

- Restate $D(M)$, S , Pi_r , and Γ_r before using the inclusion.
- Check every possible elder-rule failure mechanism is assigned to one of the two defect classes.
- Search for a self-merge or loop configuration outside the stated hypotheses.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

CL-CLS-055 — TERMINAL CLOSURE RECORD FOR EC-013

Artifact ID: CL-CLS-055-v1.0

Author line: CL — Anthropic Claude (recording clerk only; the decision is the operator's)

Created: 2026-07-21

Class: CLS — terminal closure record (proof of record for the register)

Canonical impact: EXACTLY the narrow closure below; nothing broader.

1. OPERATOR APPROVAL (verbatim provenance)

Dylan M. Roy approved the EC-013 terminal closure in the CL working session of 2026-07-21 (UTC ~21:05), replying "I approve, continue" in direct response to CL's surfaced pending-decision list, whose first item read: "EC-013 terminal approval (the one human-ready closure)". Context anchors: 00_START_HERE — EASY CLOSURE COMMAND CENTER (item 8 and the NEXT ONE-STEP section: "EC-013 remains the sole human-ready closure decision"), and HA-006's prior conditional batch, which explicitly did NOT cover EC-013. No other EC item was pending human decision, so the approval's referent is unambiguous. If the operator intends a different scope, this record is reversible per the reopening rule in section 5.

2. TERMINAL OBJECT AND LABEL (verbatim from the START_HERE package entry / GP-CLS-PROP-016)

Terminal label: CLOSED — EXACT CONDITIONAL ELDER-RULE DEFECT DICHOTOMY

Exact object: under the declared Morse, distinct-critical-value, and Morse–Smale hypotheses, $\{D(M) \neq S\}$ is contained in Π_r union Γ_r .

Critical boundary: deterministic conditional inclusion ONLY. No probability estimate, no typed-to-adjacent transfer, no R0 theorem, no P0.2, no q-rate, no Theorem B, and no machine-state conclusion is closed, implied, or strengthened by this closure.

Evidence package: GP-CLS-PROP-016-v1.0 (Drive 1PwQHKs1owsjUklKUZ1krZs6mr61eO_gPThlfFRR0BUc) and its cited sources, reviews, and negative controls — all preserved.

3. WORKFLOW EXECUTION (per the command center's post-approval steps)

- (1) Terminal object and label entered: THIS DOCUMENT is the closure record of note until the spreadsheet append lands.
- (2) Removal from the active Easy Closure Queue and (3) append of the closure row to the Closure Log (GP-REG-032 gid 200012): DELEGATED to an edit-capable line (GP) — the CL connector cannot edit spreadsheets. GP is asked to append verbatim: "EC-013 | CLOSED — EXACT CONDITIONAL ELDER-RULE DEFECT DICHOTOMY | operator approval 2026-07-21 (session relay, CL-CLS-055) | package GP-CLS-PROP-016 | scope: deterministic conditional inclusion only".
- (4) Every source, review, erratum, negative control, and reopening rule is preserved; nothing is staged or deleted by this closure.
- (5) Routine re-derivation of the exact closed object STOPS now.
- (6) Hard-front work continues without treating this narrow closure as broader proof.

4. DEPENDENCIES

Affected: the easy-closure queue (EC-013 leaves active attention); citations may now reference the terminal label instead of re-deriving the dichotomy.

Unaffected (explicit forbidden-extrapolation block): P0.1, P0.2, EC-014, Theorem B, q-machine state, adjacency probabilities, typed-vs-adjacent Palm transfer, GP-AUTO-034, all ballots and external releases.

5. REOPENING PATH

Additive only: a filed issue identifying an error in the exact object, its hypotheses, or the evidence package reopens EC-013 by a new artifact citing this record; the operator may also revoke the approval at any time, which restores EC-013 to the queue with all evidence intact.

[[OBJ:CL-CLS-055-v1.0]] [[QUEUE:EC-013]] [[OBJ:GP-CLS-PROP-016-v1.0]] [[STATUS:TERMINALLY-CLOSED]]
[[REL:RECORDS_OPERATOR_DECISION]]

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT CONDITIONAL ELDER-RULE DEFECT DICHOTOMY
Closure date	2026-07-21
Evidence basis	Deterministic pairing proof from the core pairing theorem; second-line reconstruction and terminal package.
Independent review already recorded	A strong dependent verification checked the exact inclusion without importing probability estimates.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No probability, rate, transversality, adjacency-to-one, or P0.2 conclusion.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1ZC2H_xRZa9XgaMeDnzPXffiHoemPiOIDrdvqUSQtqjg/edit

Closure / terminal record: https://docs.google.com/document/d/1ZC2H_xRZa9XgaMeDnzPXffiHoemPiOIDrdvqUSQtqjg/edit

Local source snapshot used to construct this packet: CL-CLS-055 — TERMINAL CLOSURE OF EC-013 (operator-approved 2026-07-21): CLOSED — EXACT CONDITIONAL ELDER-RULE DEFECT DICHOTOMY; register append delegated to edit-capable line

Local source snapshot SHA-256 (DOCX export, transport checksum only):
4a04eb55627b8d77290981f5926de1b986c836dadce1afc797784eca22dcd254

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.